

RE2 - Stability of a feedback amplifier: numerical solution

Data: $A_0 = 10^6$ (120 dB), $f_{p1} = 100$ kHz, $f_{p2} = 10$ MHz, $R_1 = 1$ k Ω , $R_2 = 100$ k Ω , R_L irrelevant (zero R_o).

a) Feedback gain $A(s)\beta$

$A(s) = A_0 / [(1+s/\omega_{p1})(1+s/\omega_{p2})]$, $\beta = R_1/(R_1+R_2) = 0.00990 = -40.09$ dB.

$A\beta$ mid-band = 9901 = 79.91 dB. Uncompensated: $f_u = 99.25$ MHz, PM = 5.8°, gain at 100 MHz = 0.1 dB.

f_u is the unity-gain crossover frequency, where $|A\beta| = 1$ (0 dB). The phase margin is measured there: PM = $180^\circ + \angle A\beta(f_u)$. The gain margin is read at the phase crossover, where $\angle A\beta = -180^\circ$.

b) Displacement of the first pole (PM = 45°)

Solving numerically for PM = 45° with f_{p2} fixed: $f_{p1}' = 1429$ Hz, $f_u = 10.00$ MHz, PM = 45.0°.

Gain margin: the loop has two poles, so the phase only reaches -180° at infinity (no finite phase crossover). Reading the gain at 100 MHz gives GM = 37.0 dB.

c) Insertion of an additional pole (PM = 45°)

Solving numerically for PM = 45° with f_{p1} , f_{p2} kept: $f_x = 13.9$ Hz, $f_u = 98.1$ kHz, PM = 45.0°.

Phase reaches -180° at 1.00 MHz, where GM = 37.3 dB.

Asymptotic + numerical resolution

Displaced first pole (b): asymptotic estimate $f_{p1}' = f_{p2}/(A_0\beta) = 1010$ Hz, assuming $f_u = f_{p2}$ and PM = $180^\circ - 90^\circ - 45^\circ = 45^\circ$. The asymptote ignores the -3 dB of the second pole at f_{p2} , so the exact condition $|A\beta(f_{p2})| = 1$ gives $f_{p1}' = f_{p2}/\sqrt{((A_0\beta)^2/2 - 1)} = 1428$ Hz. The numerical solution is 1429 Hz (PM = 45.00°).

Added pole (c): asymptotic estimate $f_x = f_{p1}/(A_0\beta) = 10.1$ Hz, assuming $f_u = f_{p1}$, with the second pole negligible. The numerical solution, which includes the second pole, is 13.9 Hz (PM = 45.0°).

The asymptotic values are the ones in RE2.pdf. The numerical solution includes the magnitude and phase of the second pole, so it is the value that meets PM = 45° exactly.

d) Comparison

Added pole: simple to implement (one RC network), but the crossover falls to about 98 kHz, so the bandwidth is lower. Displaced first pole: crossover at 10 MHz, i.e. a much larger bandwidth, but it needs a dominant-pole network that is harder to build. The gain margin is 37 dB for the added pole and is not finite for the displaced pole.

Errors found in the original resolution (RE2.pdf)

- $\beta = 0.1$ is printed, but $R_1/(R_1+R_2) = 1/101 = 0.0099$. The printed -40 dB is correct, the value 0.1 (-20 dB) is not.

- **Gain margin 40 dB at 100 MHz** is not a true margin: the phase is not -180° there (it is about -174° for the displaced pole). The gain at 100 MHz is 37.0 dB for the displaced pole. Only the added-pole case has a finite phase crossover, at 1.00 MHz, with GM = 37.3 dB.